

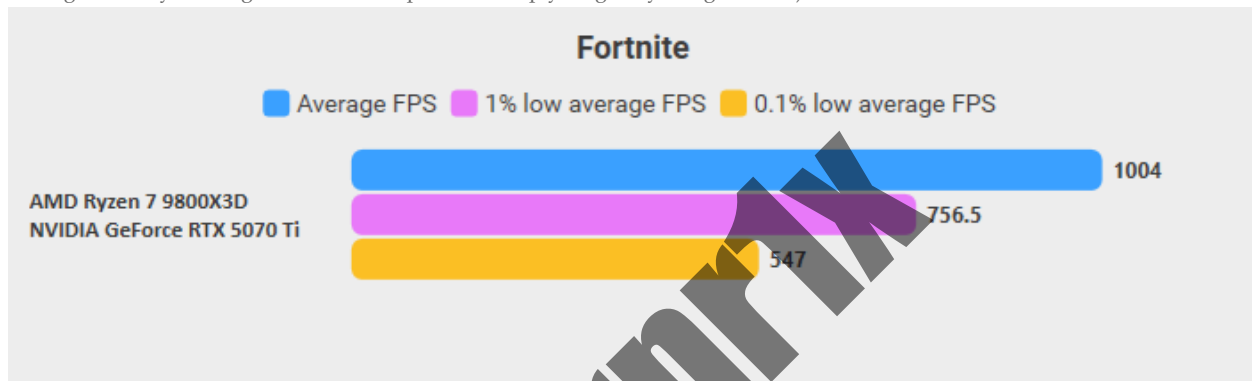
Turnr1x AM5 X3D Bios Guide

Overview

This is a guide for AM5 X3D bios settings that I have concluded provide the best performance. For my specific case I am on an X3D (9800X3D) chip and these settings yielded the best results for my system. While this guide works perfectly fine in 99.99% of cases, **DO NOT** mass apply the settings, I do not recommend it. Each setting should be tested independently on your system to make sure they provide better performance and do not affect your system negatively in any way. I **am not liable** for any damage done to your system.

Benchmark Results

(These are placed at the top just to show these settings do in fact have a positive impact. Do not DM me saying “I get 1200fps looking at the sky dancing in a creative map”, I will simply laugh at your ignorance.)



(My results from the Fortnite minecart benchmark at 1080p. Results are the aggregate of 5 total runs. I do not have my old picture from CS2 bench, but P1 lows were slightly over 400 with bots at 1280x960.)

Settings

ACP Power Gating = Disabled
ACP Clock Gating = Disabled
Unused GPP Clocks Off = Disabled
PME Turn Off Support = Disabled
S3 PCIe Save Restore Mode = Disabled or Mode 2 (TEST)
Global C-state Control = Disabled
DF Cstates = Disabled
Disable DF to external downstream IP Sync Flood Propagation = Disabled
Disable DF Sync flood propagation = Disabled
Freeze DF module queues on error = Disabled
PSPP Policy = Disabled
Core Watchdog Timer = Disabled
Core Watchdog Timer Severity = No error
Power Supply Control = Typical Current Idle
Prochot VRM Throttling = Disabled
ACPI_CST C1 Declaration = Disabled
ACPI Standby State = Disabled
ASPM Control for CPU PCIe = Disabled

PAUSE Delay = Disabled or Auto (TEST)
Streaming Stores Control = Enabled
OPCache Control = Enabled
FCH Spread Spectrum = Disabled
Core Tuning Configuration for Gaming = Level 2 or Legacy (Level 2 yielded the best results for me)
SVM Enable = Disabled
IOMMU = Disabled (UNLESS NEEDED FOR CERTAIN GAME REQUIREMENTS)
REP-MOV/STOS Streaming = Enabled
Enhanced REP MOVSB/STOSB = Enabled
Fast Short REP MOVSB = Enabled
L3 DFLS Stretch Mode = Disabled
L1/L2 Stream HW Prefetcher = Enabled
L1 Stride Prefetcher = Enabled
L1 Region Prefetcher = Enabled
L1 Burst Prefetch Mode = Enabled
L2 Up-Down Prefetcher = Enabled
CPU Speculative Store Modes = More Speculative
PSS Support = Disabled
PS2 Devices Support = Disabled
ACPI SRAT L3 Cache As Numa Domain = Disabled
Latency Under Load = Disabled or Auto (TEST)
iGPU/Integrated Graphics = Disabled
Any Motherboard Driver Installer/ App = Disabled
Discrete GPU's USB Port = Disabled
USB power delivery in Soft Off state (S5) = Disabled
TSME = Disabled
SMEE = Disabled
Data scramble = Disabled
Memory P-state = Disabled
Memory Fast Boot = Disabled
Memory Context Restore = Disabled
Power Down = Disabled
ECC = Disabled
PCIe All Port ECRC = Disabled
PCIe loopback Mode = Disabled
Persistence mode for legacy endpoints = Disabled
NPU Deep Sleep Enable = Disabled
Advanced Error Reporting = Disabled
Platform First Error Handling = Disabled
MCA Error Thresh = False
MCA FruText = False
SMU and PSP Debug Mode = Disabled
Action on BIST Failure = Dp nothing
Log Transparent Errors = Disabled
BankSwapMode = Swap APU
MBIST Enable = Disabled
MBIST Aggressors = Disabled
MBIST Per Bit Slave Die Reporting = Disabled
Aggressor Channel = Disabled
Onboard Wifi = Disabled (Unless you use it)
Onboard Bluetooth = Disabled (Unless you use it)

—Disable ALL of the below D3 settings—

ACPI D3 Support
Internal PCIe GPP 0 D3
SOC GPU D3
SOC HD AUDIO D3
SOC AZALIA D3
SOC USB 3.1 D3
SOC ACP D3
SOC USB 2.0 D3
SOC USB 3.1 for USB 4 D3
SOC USB D3
Internal USB4 PCIe Tunneling D3

Conclusion

THANK YOU for checking out my guide, I greatly appreciate it! I hope it has provided your system with better performance and made gaming more enjoyable. There are obviously many more settings that can be changed to increase performance but a lot are system dependent and require too much testing to just slap on a guide and call it "universal". If you like this drop a follow on my socials @Turnr1x on pretty much everything you can think of. If this gets the right support I hope to do many more things like this in the future. If you have any questions or want some more advice, reach out to me on discord or twitter. Thanks again!

Turnr1x